The worldline action and uniform local Barnes asymptotics in the recoil limit

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Working research note — 30 October 2026

Abstract

We supply an analytic action identity for the reflected recoil Barnes saddle at general positive Euclidean tetrahedron masses. An elementary quadratic-field identity identifies each logarithmic mass derivative with the corresponding exterior dihedral angle; homogeneity then gives the worldline action. We also establish uniform estimates for the actual localized Gamma quotient, retaining the transverse energies. These estimates justify the integrated removal of its exact periodic pole factor even when the pole-row separation tends to zero. The result concerns a specified local contribution. Dominance in the full Barnes cycle, transverse tails, and the physical four-point dictionary remain open.

1 Definitions and the order of limits

The de Sitter radius is one. We first take the fixed-mass $\kappa \to \infty$ limit, and only afterwards increase the masses. This does not assert a joint finite-G limit. The object remains the internal chiral amplitude of the fixed-topology CLP graph [2]; no new physical state or external energy cycle is specified.

For edge labels $\mu_1, \dots, \mu_6$, let a be the four half-sums on $123, 156, 246, 345$, and r the three half-sums on $1245, 1346, 2356$. The exact reflected kernel is

$$\mathcal{B}(z) = H(z)E(z), \tag{1}$$

$$H(z) = 8\pi^3 e^{2\pi i z - \pi \sum r} \frac{\prod_a \Gamma(z + ia)}{\Gamma(z + \frac{1}{2}) \prod_r \Gamma(z + \frac{1}{2} + ir)}, \qquad E(z) = \prod_r (1 + e^{2\pi i (z + ir)})^{-1}. \tag{2}$$

Use $f(t) = t(\log t - 1)$ and

$$\Phi(z) = 2\pi i z - \pi \sum r + \sum_a f(z + ia) - \sum_r f(z + ir) - f(z), \tag{3}$$

$$g(z) = 8\pi^3 \prod_a (z + ia)^{-1/2}. \tag{4}$$

The branches below are fixed by the right saddle. With $s = iz$,

$$P(s) = \prod_a (s - a) - s \prod_r (s - r) = A_q s^2 + Bs + C, \qquad \Delta = B^2 - 4A_q C = -9V_{\text{dual}}^2.$$

Here A_q is a quadratic coefficient, distinct from the large parameter K used later.

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2 An action identity for all six real masses

Proposition 1. *For a positive, nondegenerate Euclidean dual tetrahedron, set* $T = \sqrt{-\Delta} = 3V_{\mathrm{dual}} > 0$, $s_+ = (-B + iT)/(2A_q)$, *and* $z_+ = -is_+$. *Then, with the indicated logarithm branches,*

$$-2\Re\Phi(z_+) = \sum_{j=1}^{6} \mu_j d_j = I_{\mathrm{CSTY}}, \tag{5}$$

where $d_j \in (0, \pi)$ *are the exterior dihedral angles in CSTY Appendix B.1 [1].*

Proof. Write $\boldsymbol{\mu} = (l, b, c, f, d, e)$. The four faces are $(l, b, c), (l, d, e), (b, f, e), (c, f, d)$. The edge opposite l is f, and $A_q = (lf + bd + ce)/2 > 0$, so $\Re z_+ > 0$. Define

$$D_l = [l^2 - (b - c)^2][l^2 - (d - e)^2] > 0, \tag{6}$$

$$H_l = (l^2 + b^2 - c^2)(l^2 + e^2 - d^2) - 2l^2(b^2 + e^2 - f^2). \tag{7}$$

Let

$$a_1 = \frac{l + b + c}{2}, \quad a_2 = \frac{l + d + e}{2}, \quad r_1 = \frac{l + b + f + d}{2}, \quad r_2 = \frac{l + c + f + e}{2}.$$

The quadratic relation gives

$$R_l := \frac{(z_+ + ia_1)(z_+ + ia_2)}{(z_+ + ir_1)(z_+ + ir_2)} = \frac{H_l + 8ilT}{D_l}. \tag{8}$$

This can be checked without radicals: replace iT by $2A_q s + B$ and cross-multiply; the resulting polynomial is divisible by $P(s)$. The new code verifies the zero remainder with all six masses symbolic. The further polynomial identity

$$H_l^2 + 64l^2T^2 = [16\,\mathcal{A}(l, b, c)\mathcal{A}(l, d, e)]^2$$

uses the Heron area $\mathcal{A}$. Since $D_l, l, T > 0$, R_l lies in the upper half-plane. Its argument has exactly the cosine specified by CSTY (B.6), hence $\arg R_l = d_l$.

All arguments $z_+ + ia, z_+ + ir, z_+$ lie in the right half-plane. The exponentiated saddle equation therefore makes $\Phi'(z_+)$ a continuous integer multiple of $2\pi i$. The domain of nondegenerate Euclidean edge lengths is connected: squared edge lengths are linear functions of the positive definite Gram matrix of the three edges at a chosen vertex. At the regular tetrahedron the integer is zero, so $\Phi'(z_+) = 0$ throughout this domain.

Differentiating on shell now gives

$$\partial_l[-2\Re\Phi(z_+)] = 2\pi + \Im\left[\log(z_+ + ia_1) + \log(z_+ + ia_2) - \log(z_+ + ir_1) - \log(z_+ + ir_2)\right] = d_l. \tag{9}$$

The last branch integer is again fixed at the regular tetrahedron and cannot jump on the connected domain. The other five edges follow by symmetry. Finally, $\sum a = \sum r$ cancels every scaling term $\lambda \log \lambda$; the on-shell potential is homogeneous of degree one. Euler's identity gives (5). □

At complex masses, continue the two chiral functions separately:

$$\Phi^\dagger(\boldsymbol{\mu}) = \overline{\Phi(\overline{\boldsymbol{\mu}})}, \qquad I = -\Phi - \Phi^\dagger.$$

The equality extends in any common analytic continuation domain with the transported angle and saddle branches. It does not mean taking a new modulus at arbitrary complex masses, and it does not determine which saddle contributes to an integration cycle.

3 Uniform estimates for the actual local amplitude

Use the equal-probe labels

$$(\mu_1,\dots,\mu_6)=(m_1,\nu,m_2,m_3,\nu,m_4),\quad m_{1,3}=M+\omega/2\pm u,\quad m_{2,4}=M-\omega/2\pm v.$$

We take the joint large-mass regime

$$K=\frac{\nu^2}{2M}\to\infty,\quad \epsilon=\frac{\nu}{M}\to 0,\quad \omega=iKy,\quad |y|\le Y,\quad \rho=\frac{|u|+|v|}{\nu}\to 0,\qquad u,v\in\mathbb{R}. \tag{10}$$

This includes transverse boxes $|u|,|v|\le L\sqrt{M}$, $L=o(\sqrt{K})$. One can impose the earlier WKB window $\nu^{4/3}\ll M\ll\nu^2$ in addition. The local result below does not extend that separate WKB theorem.

Write $z_+=x_+-ic$. Direct quadratic algebra gives

$$\delta:=M+\nu-c=\frac{\nu(\nu^2-u^2-v^2+\omega^2)}{4M^2+2\nu^2-2u^2-2v^2+\omega^2}\sim\frac{\nu^3}{4M^2}>0,\qquad \frac{x_+}{K}=\frac{\sqrt{1+y^2}}{2}+O(\epsilon^2+\rho^2). \tag{11}$$

Choose a fixed

$$0<\eta<\frac{\sqrt{1+Y^2}-Y}{4}.$$

For real $|h|\le\eta K$, the two near denominator parameters are

$$z_++h+i(M+\nu\pm\omega/2)=x_++h\mp Ky/2+i\delta,$$

whose real parts exceed $c_Y K$. The four numerator arguments have modulus comparable to ν; the two other denominator arguments have modulus comparable to M. All stay in a fixed sector away from the negative real axis. The left pole rows remain above the line and the right rows below. Thus the horizontal contour can follow this imaginary-energy path without pole crossing. This finite-parameter statement is not a global asymptotic estimate.

Let χ be a fixed even C^4 cutoff, $0\le\chi\le 1$, equal to one for $|t|\le 1/2$ and zero for $|t|\ge 1$, and define

$$F_K(x)=\chi\left(\frac{x-x_+}{\eta K}\right)\frac{H(x-ic)}{g(z_+)e^{\Phi(z_+)}}. \tag{12}$$

Theorem 1 (Localized Gamma quotient)**.** *In* (10)*, uniformly in the stated parameter sets,*

$$\|F_K^{(4)}\|_1\le CK^{-3/2}, \tag{13}$$

$$\int_{\mathbb{R}}F_K(x)\,\mathrm{d}x=\sqrt{\frac{2\pi K}{q_*}}\,[1+O(K^{-1})],\qquad q_*=-K\Phi''(z_+)=4\sqrt{1+y^2}+O(\epsilon+\rho^2). \tag{14}$$

The square root is chosen with $\Re q_*>0$. In particular the integral has modulus at least $c\sqrt{K}$.

Proof. The sector and modulus bounds permit the fixed-shift Stirling expansion with remainder [3]:

$$H(z)=g(z)e^{\Phi(z)+R(z)},\qquad R^{(j)}(z)=O(K^{-1-j})\quad(0\le j\le 4).$$

For each local point, a complex disk of radius $c'K$ stays in a safe sector for all Gamma arguments. The scalar $O(K^{-1})$ remainder and Cauchy's estimates give the derivative bounds. The half-integer shifts in (2) are already included in g, and

$$(\log g)^{(j)}=O(\nu^{-j})\quad(1\le j\le 4).$$

Direct differentiation also gives $\Phi^{(j)} = O(K^{1-j})$, $2 \le j \le 6$. The two near denominators contribute $-1/(z+ir_+) - 1/(z+ir_-)$ to Φ''. Their real part is at most $-c/K$; all other terms are $O(\nu^{-1}) = o(K^{-1})$. Consequently throughout the local segment

$$\Re[\Phi(z_+ + h) - \Phi(z_+)] \le -ch^2/(2K).$$

Here stationarity is exact, by continuation of $\Phi' = 0$.

Apply the fourth exponential derivative formula to $\Phi + \log g + R$, then differentiate the cutoff. With constants adjusted as necessary, the result is bounded by

$$CK^{-2}(1 + |h|/\sqrt{K})^4 e^{-c'h^2/K} + CK^{-4}e^{-c''K}\mathbf{1}_{\{\eta K/2 \le |h| \le \eta K\}}.$$

Integration proves (13). The cutoff terms have not been dropped; their support is exponentially suppressed.

For the integral, put $h = \sqrt{K}\,t$. The quadratic exponent is $-q_* t^2/2$. The cubic phase and the linear amplitude correction are odd polynomials of size $O(K^{-1/2})$, so their integrals against the even complex Gaussian and even cutoff vanish. Fourth-order terms, the remainder R, and squares of the odd corrections give $O(K^{-1})$. For a uniform justification first expand on $|t| \le K^{1/12}$; the preceding concavity estimate controls its exterior. Finally q_* lies in a fixed compact subset of the right half-plane and has the displayed limit, as follows from $\Phi'' = -\sqrt{-\Delta}/\prod_a (z_+ + ia)$. The Gaussian integral is bounded away from zero, proving the relative estimate and lower bound. □

4 The pole factor is now controlled inside the local integral

The preceding note proved the following elementary estimate: for $|q_r| < 1$, $E_q(x) = \prod_{r=1}^{3}(1 + q_r e^{2\pi i x})^{-1}$ and $F \in W^{4,1}(\mathbb{R})$,

$$\left| \int F(E_q - 1) \right| \le C_4 \|F^{(4)}\|_1, \qquad C_4 = \frac{\zeta(2) + 3\zeta(3) + 2\zeta(4)}{2(2\pi)^4}. \tag{15}$$

Indeed the n-th Fourier coefficient is bounded by $\binom{n+2}{2}$; four integrations by parts give a summable bound. The constant is independent of how close the q_r's are to the unit circle, although the endpoint itself is excluded.

On our line $q_r = e^{-2\pi(r-c)}$ and $|q_r| < 1$. The theorem verifies both hypotheses needed to turn (15) into the relative estimate

$$\frac{|\int F_K(E-1)|}{|\int F_K|} = O(K^{-2}), \tag{16}$$

uniformly even when $\delta \downarrow 0$. Thus the actual local contribution

$$\mathfrak{B}_{\mathrm{loc}} = \frac{1}{2\pi i}\int_{\mathbb{R}} \chi\left(\frac{x - x_+}{\eta K}\right) \mathcal{B}(x - ic)\,\mathrm{d}x$$

obeys

$$\mathfrak{B}_{\mathrm{loc}} = \frac{g(z_+)e^{\Phi(z_+)}}{i\sqrt{2\pi[-\Phi''(z_+)]}}\,[1 + O(K^{-1})]. \tag{17}$$

The orientation is left to right; the Gaussian branch is fixed by $\Re[-\Phi''] > 0$. This establishes the local expression using the actual Gamma quotient and pole factor. It does not replace $\mathfrak{B}_{\mathrm{loc}}$ by the full $\mathfrak{B}$. Real ω away from zero is also outside the uniform regime of this theorem: its proposed saddle line can cross pole rows.

5 Diagnostics and the remaining global problem

The script `code/recoil_local.py` records 50 passing checks: general symbolic angle, modulus and Cayley–Menger identities, the positive quadratic coefficient and transverse pole height, ten cutoff jets, all six angles on three unequal tetrahedra, two separate-chiral complex continuations, and refined local quadratures. The proofs above, rather than finite-point sampling, support the uniform claims.

For the local calculation take $\epsilon = K^{-3/2}$, $u = .3\sqrt{M}$, $v = -.2\sqrt{M}$, and $\eta = .1$. The cutoff transition is the degree-nine polynomial $1 - 126s^5 + 420s^6 - 540s^7 + 315s^8 - 70s^9$, with $s = 2|h|/(\eta K) - 1$ on its transition interval. It has the required four vanishing endpoint jets. The actual Gamma quotient, rather than its Gaussian approximation, is integrated.

K	y	pole height δ	relative Gaussian error
400	0	2.4996×10^{-2}	3.7970×10^{-3}
1600	1	1.2499×10^{-2}	9.1941×10^{-6}
6400	1	6.2499×10^{-3}	2.3011×10^{-6}

Quadrature is refined from 48 to 96 nodes per segment at 45/60 digits. The complex integral is separately repeated at 80 digits on the same 96-node segmentation. The fourth-derivative L^1 norm has nearly sharp absolute-value corners. The initial unsplit grid failed its fixed tolerance; splitting at zeros of the actual derivative's real part resolved it. Both the initial code and failed output are retained. These are empirical diagnostics, not certified interval errors. We have not directly quadrature-tested the complete large-mass integral with E; its local control follows from (15)–(16).

The next required result is a relative bound on $\mathfrak{B} - \mathfrak{B}_{\mathrm{loc}}$, including the other saddle, the negative-real direction and both remote tails. Transverse exterior regions, physical insertion factors, the three external energy cycles and disconnected baseline remain separate requirements. The work does not establish finite-G dynamics, a microscopic state dictionary, or a complete de Sitter dual. This is an author self-reviewed proof draft without independent certification or a priority claim.

References

[1] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1, Appendix B.1.

[2] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1.

[3] NIST DLMF, Section 5.11, fixed-sector Stirling expansions, shifted forms and remainder bounds.